

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP

#XUVV♥ ORGANIC CHEMISTRY: SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-08 CHEMISTRY

VISHAL SIR



JK Bole Wallah

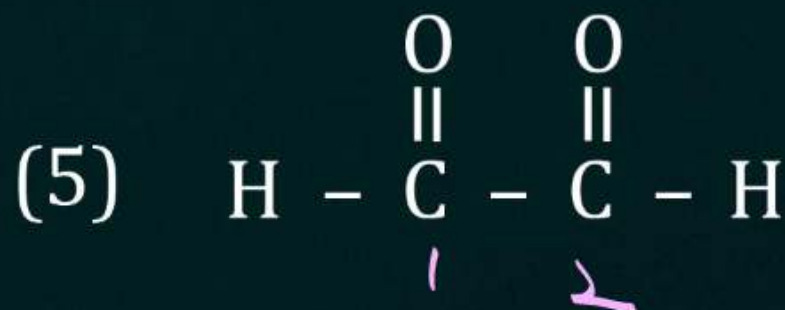
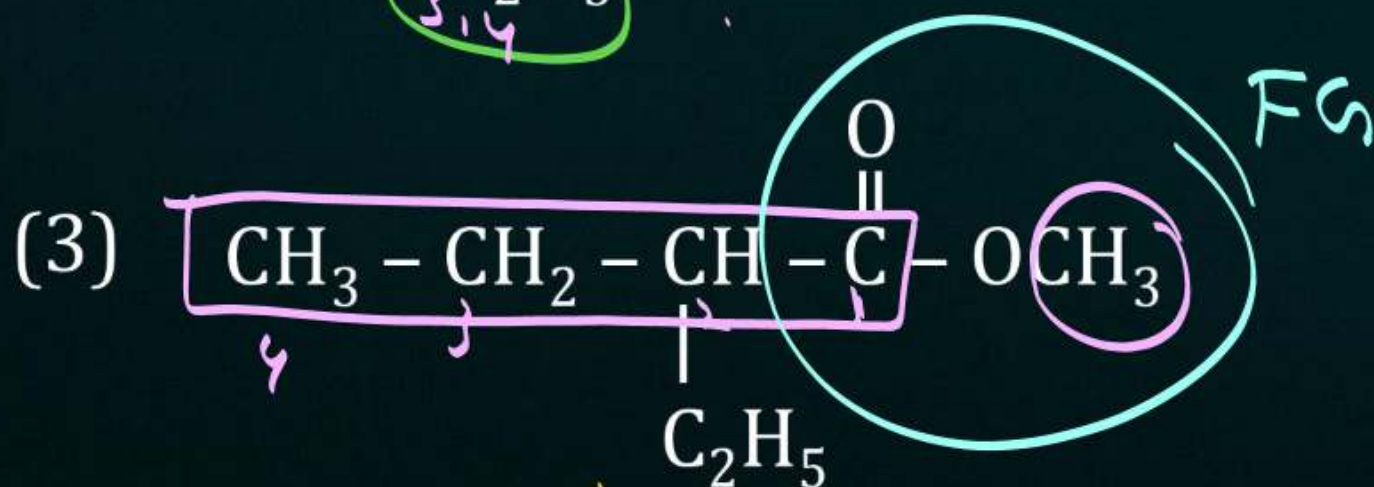
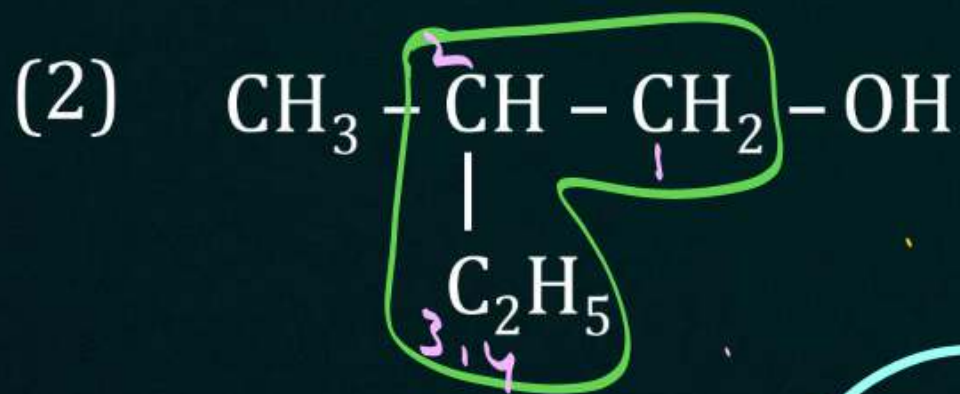
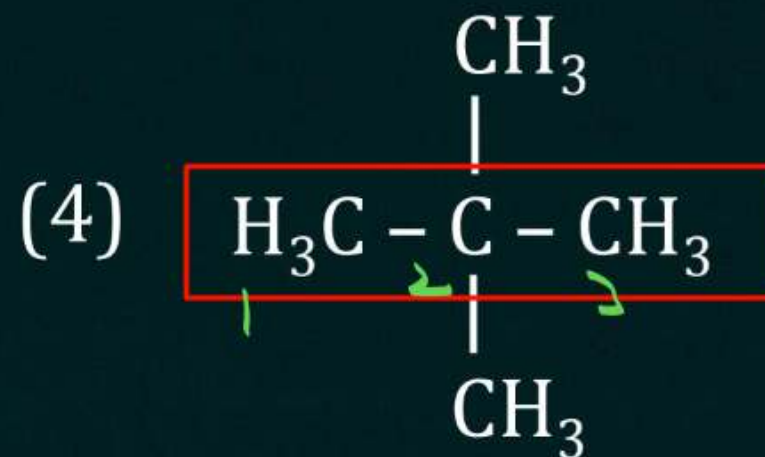
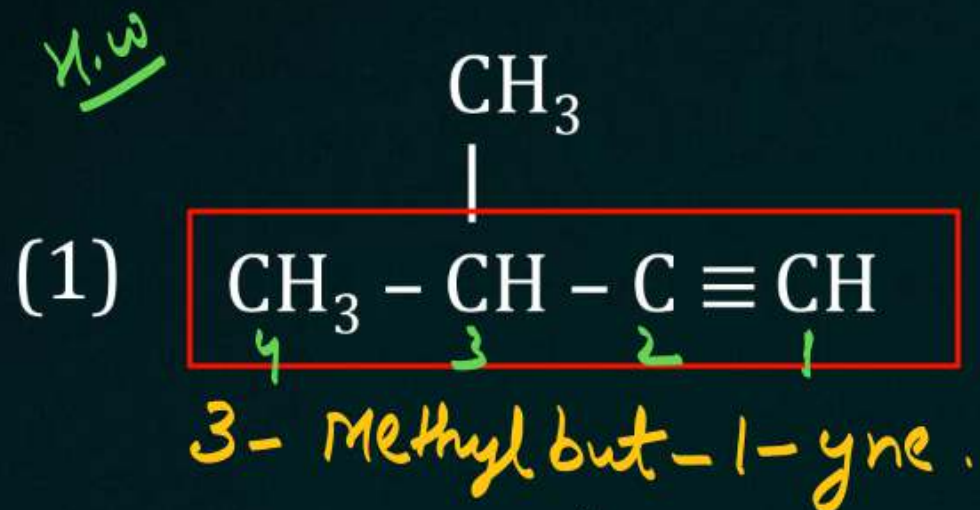
Topics To Covered

- 1** Nomenclature of Functional Groups Part 3
- 2** Electron Displacement Effects

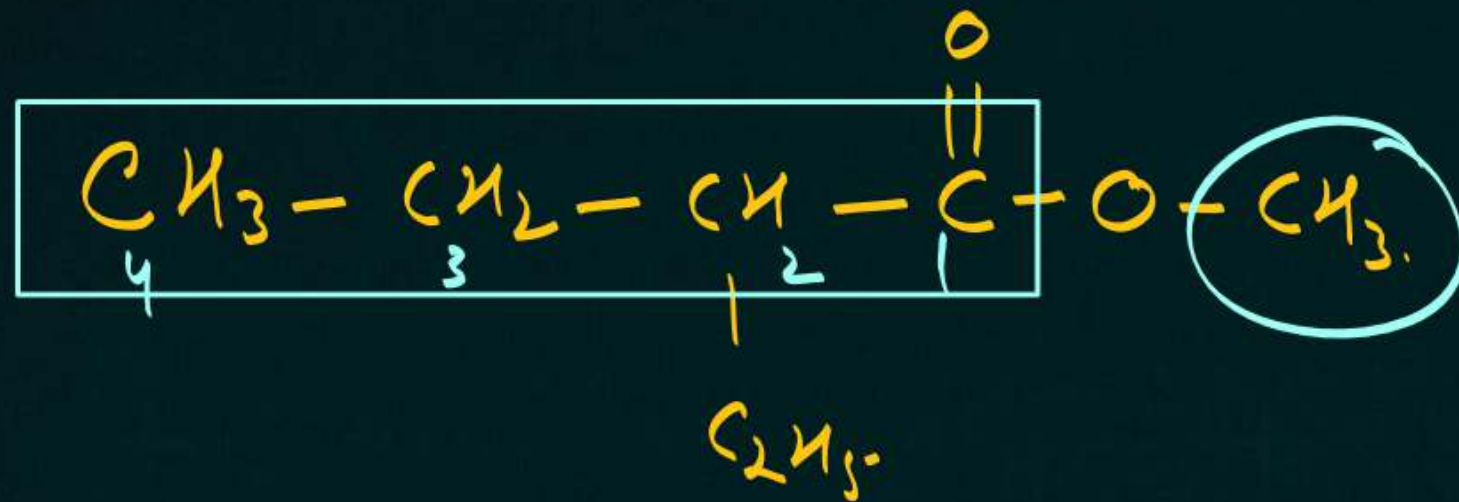
QUESTION

Write the IUPAC names of following compounds.

[2025]



FG > MB

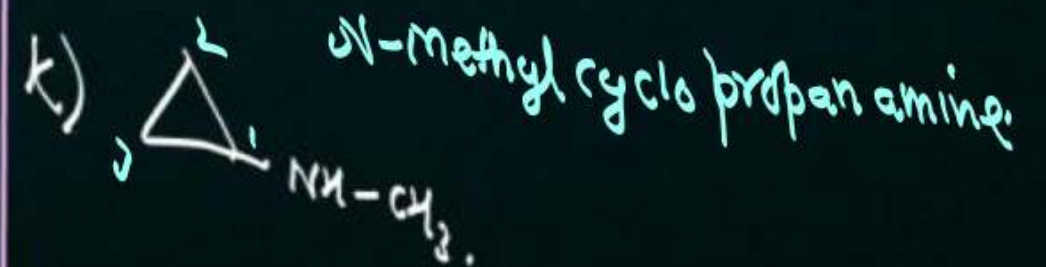
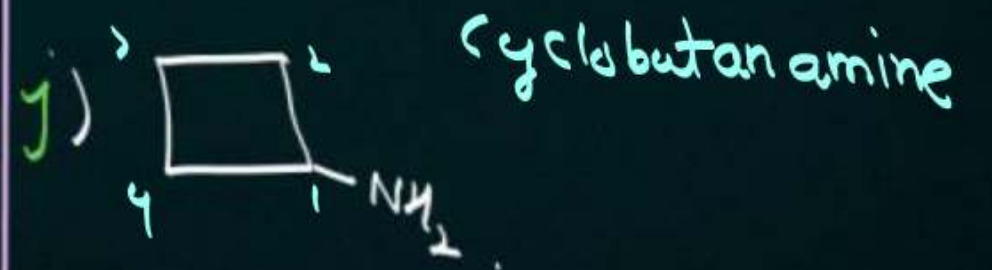
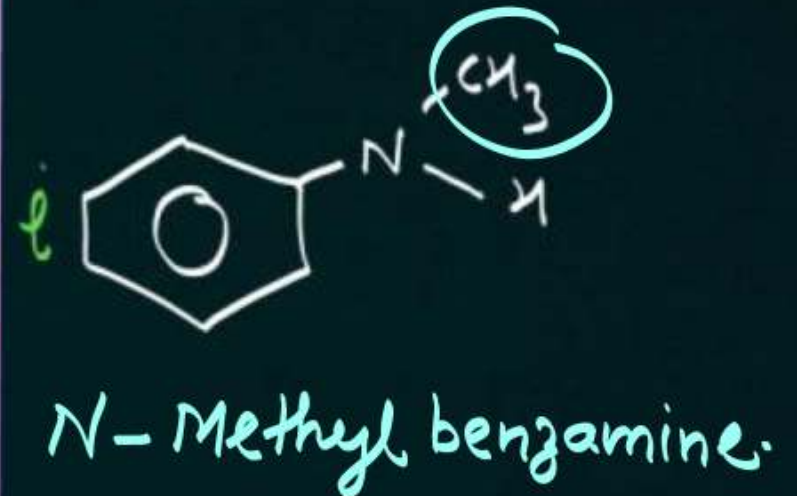
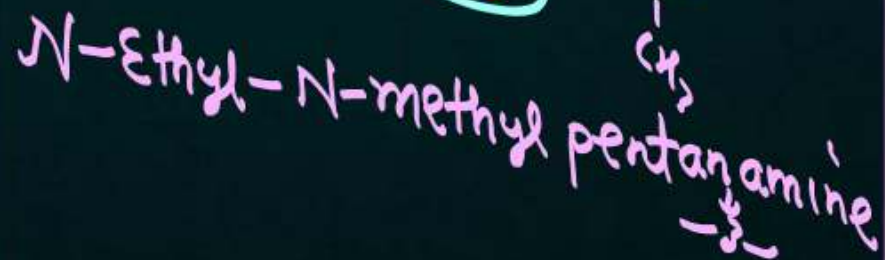
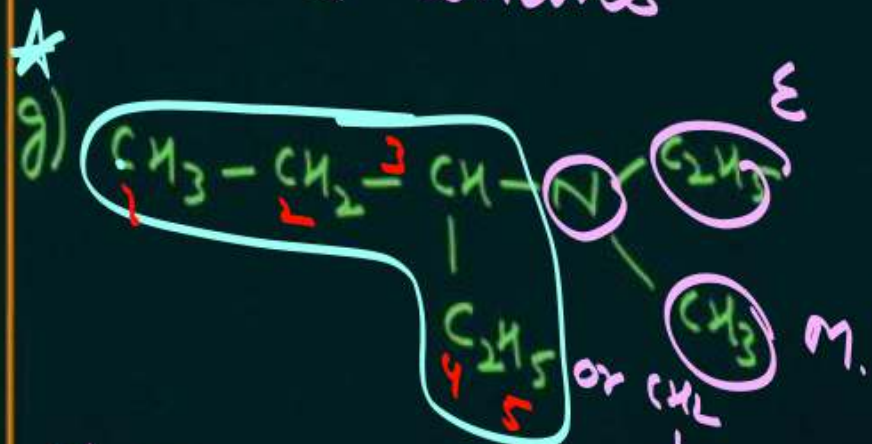
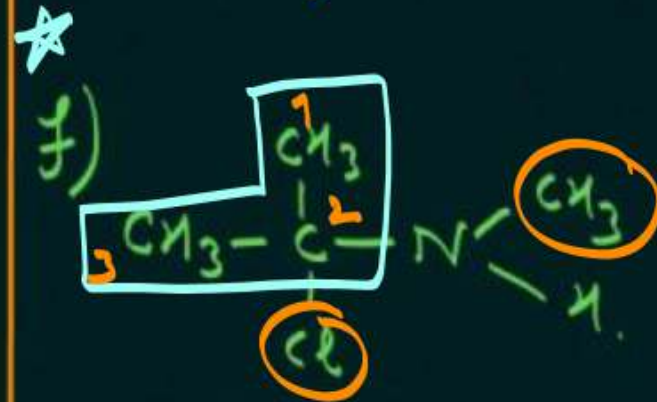
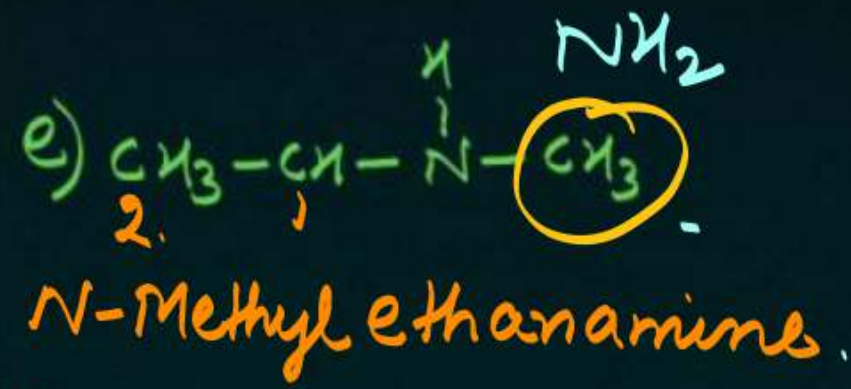
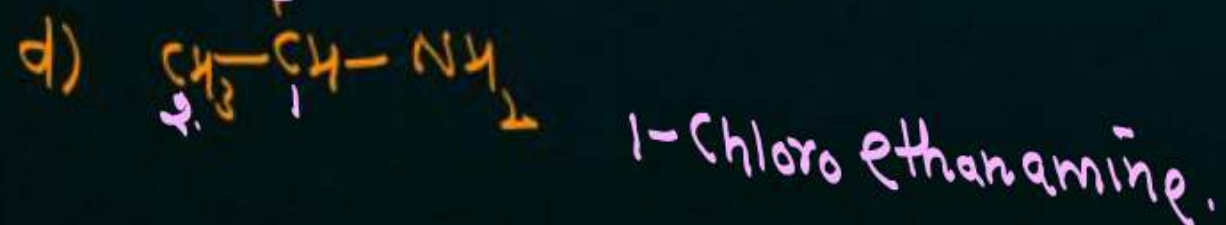
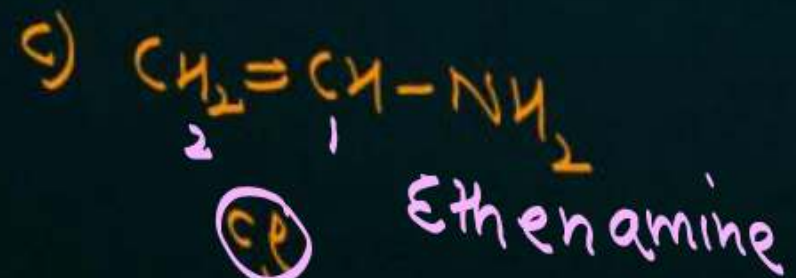
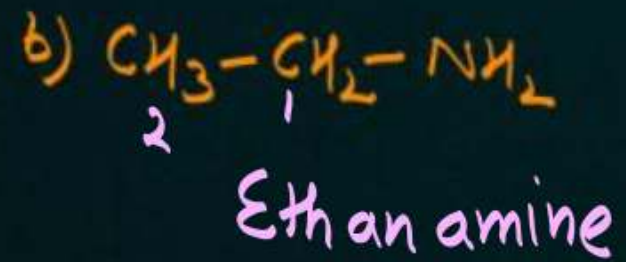
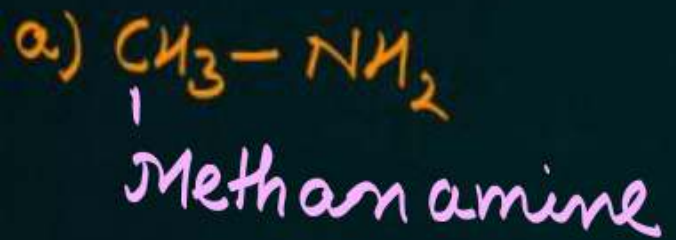


Methyl-2-ethyl butan-1-oate.

9) AMINES $\rightarrow$ $(-NH_2)$ $R-NH_2$.

2° - Suffix = amine

Subs + Cyclo + WR + ane or ene or yne + amine

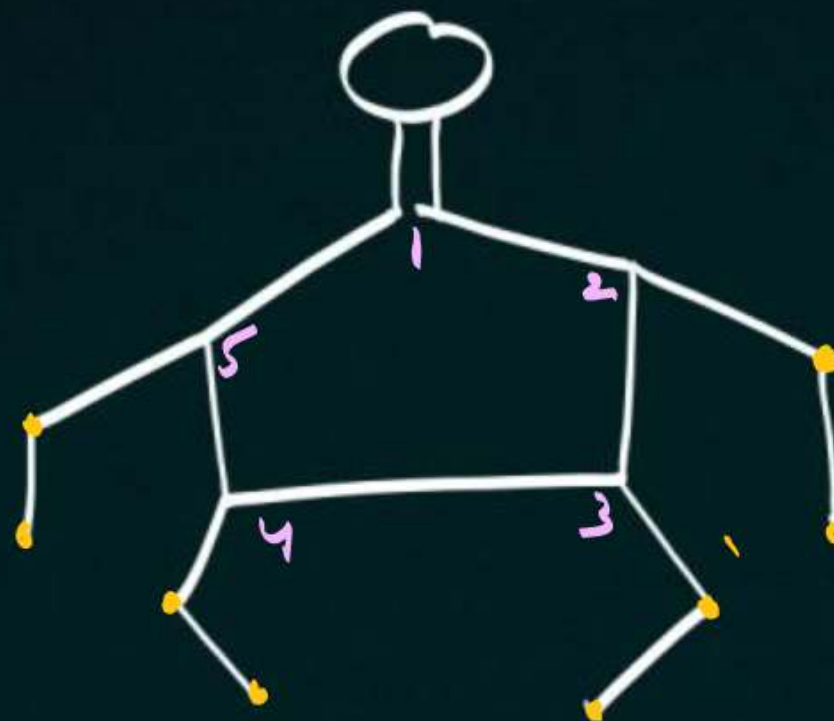


PRACTICE →

PCC → MD



5-Cyclopropylcyclopenta-1,3-diene



2,3,4,5-Tetraethyl
cyclopentan-1-one



5-(cyclohex-2'-enyl)-2,3-diethyl
cyclohexan-1-ol.

NOMENCLATURE OF COMPOUNDS CONTAINING MORE THAN ONE FUNCTIONAL GROUP.

1) $\text{—}\overset{\text{O}}{\parallel}\text{C}\text{—OH}$ (carboxylic acid)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ oic acid	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ Carboxy.
2) $\text{—}\overset{\text{O}}{\parallel}\text{C}\text{—OR}$ (ester)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ oate	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ Alkoxycarbonyl.
3) $\text{—}\overset{\text{O}}{\parallel}\text{C}\text{—X}$ (acid halide)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ oyl halide	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ Halocarbonyl.
4) $\text{—}\overset{\text{O}}{\parallel}\text{C}\text{—NH}_2$ (acid amide)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ amide	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ Carbamoyl.
5) $\text{—C}\equiv\text{N}$ (cyanides)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ nitrile	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ Cyano
6) —NC (Isocyanides)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ isonitrile	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ Isocyano.
7) $\text{—}\overset{\text{O}}{\parallel}\text{C}\text{—H}$ (aldehyde)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ al	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ Oxo or Formyl
8) $\text{C}\text{—}\overset{\text{O}}{\parallel}\text{C}\text{—C}$ (ketone)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ one	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ oxo.
9) —OH (alcohol)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ ol.	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ hydroxy.
10) —NH_2 (amine)	$\xrightarrow{\text{2}^\circ \text{ suffix}}$ amine	$\xrightarrow{\text{2}^\circ \text{ Prefix (substituent)}}$ amino.

If a molecule has more than one functional group then only one functional group is taken as senior functional group & others are treated as substituent.

PRIORITY ORDER FOR CHOOSING SENIOR FUNCTIONAL GROUP $\Rightarrow$



NOTE $\Rightarrow$ $-COOH$, $-CHO$, $-C(=O)-$ when acts as substituent their carbon may or may not be included in PCC depending upon the priority order.

Note $\Rightarrow$ In case of $-CHO$ $\begin{cases} \rightarrow 2^\circ \text{ prefix} - \text{oxo} \rightarrow \text{when included in PCC} \\ \rightarrow 2^\circ \text{ prefix} \rightarrow \text{Formyl} \rightarrow \text{when not included in PCC} \end{cases}$

— COOH ✓

— COOR ✓

— COX ✓

— CONH₂ ✓

— CN ✓

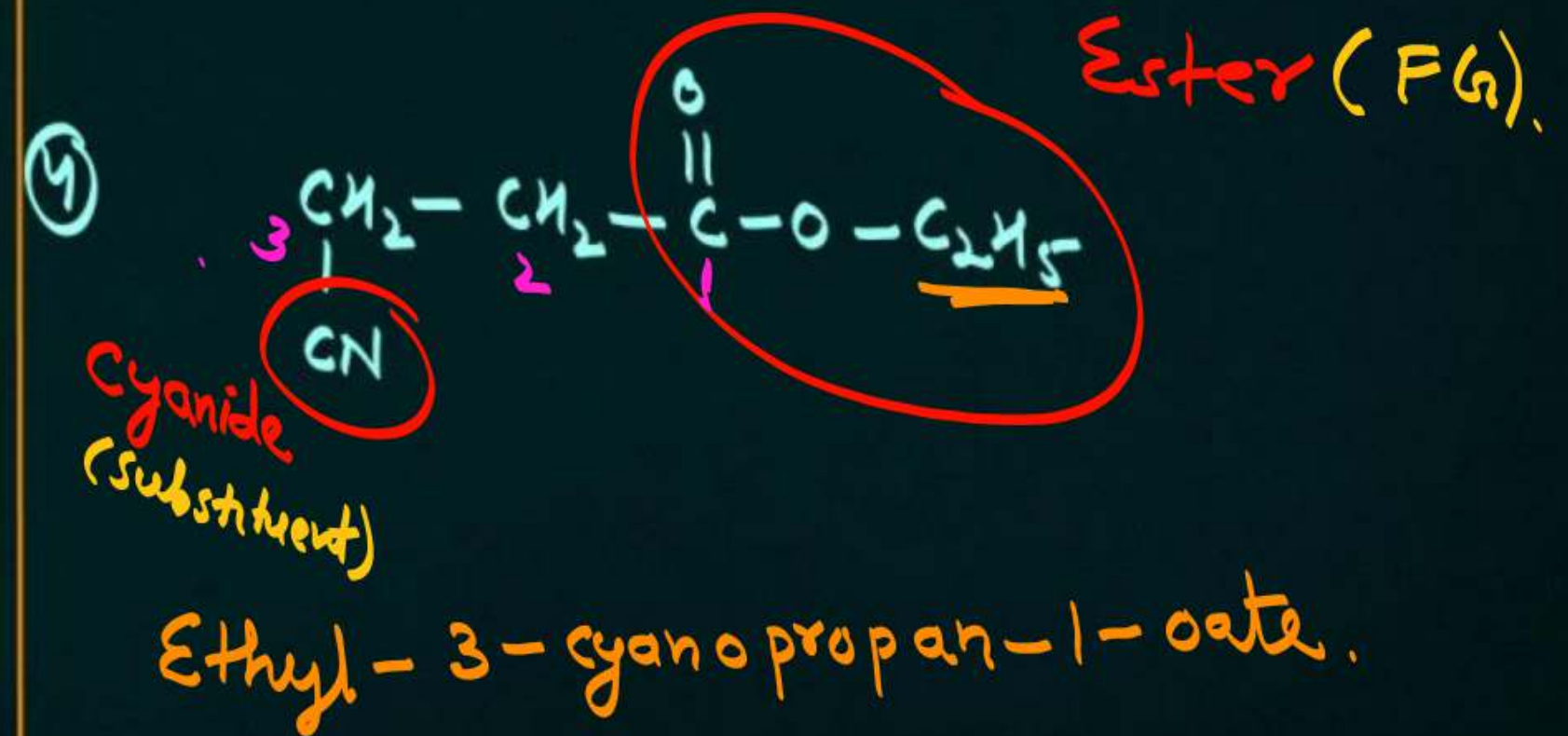
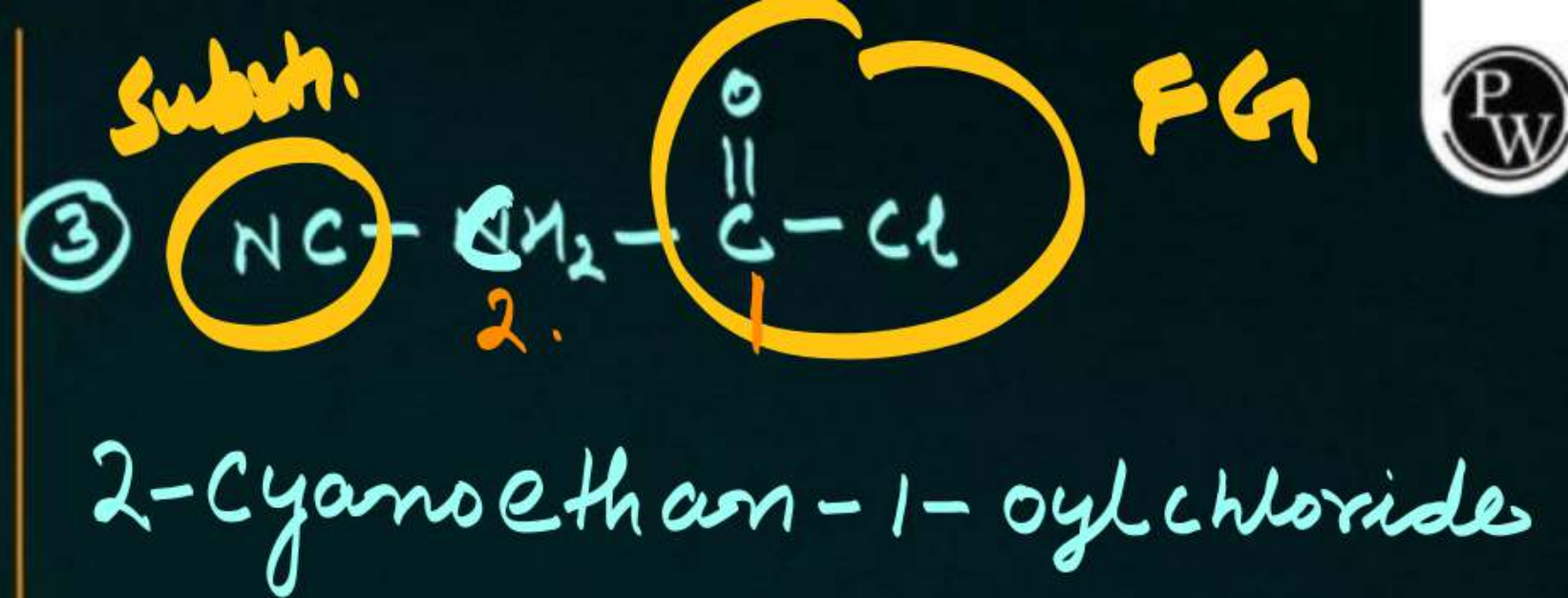
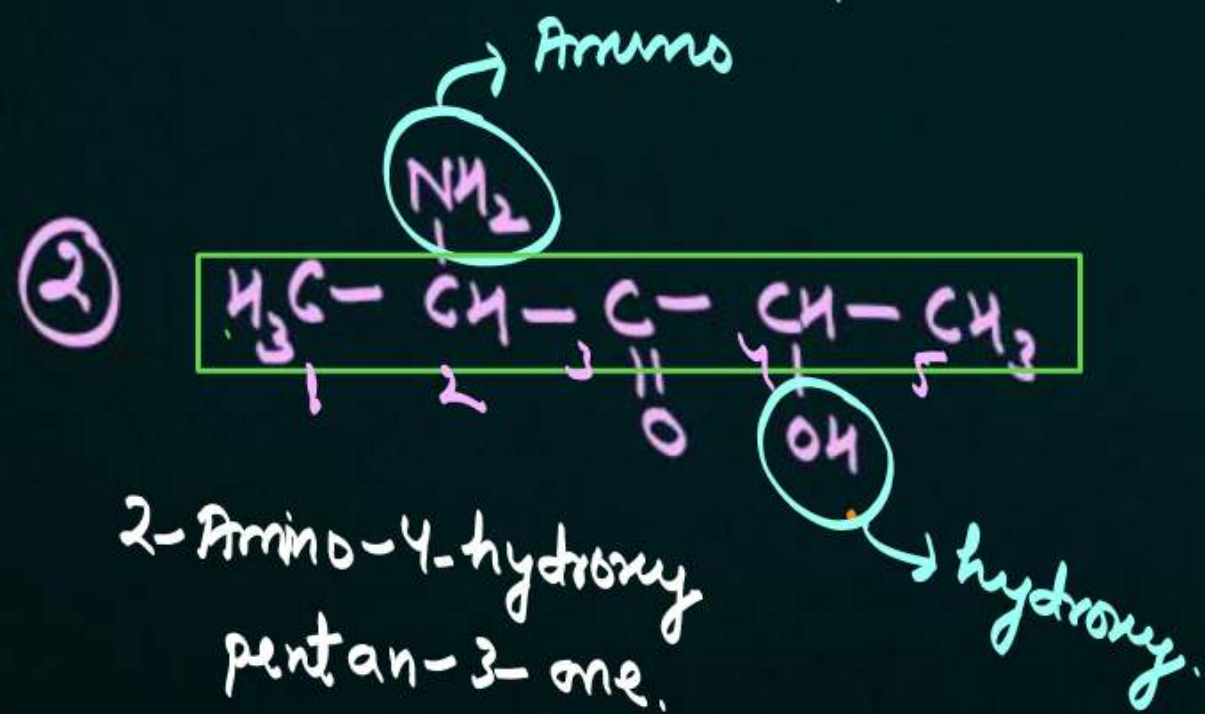
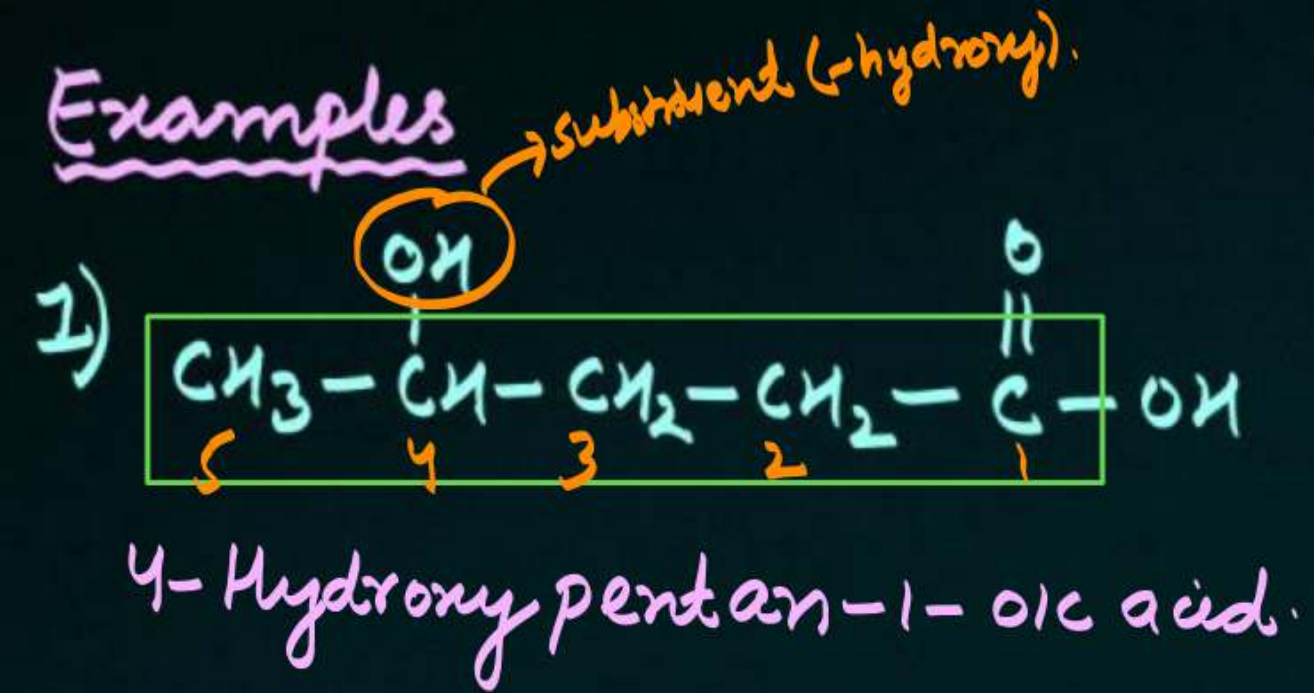
— NC ✓

— —CHO ✓

— $\begin{array}{c} \text{O} \\ \parallel \\ \text{C} - \end{array}$ ✓

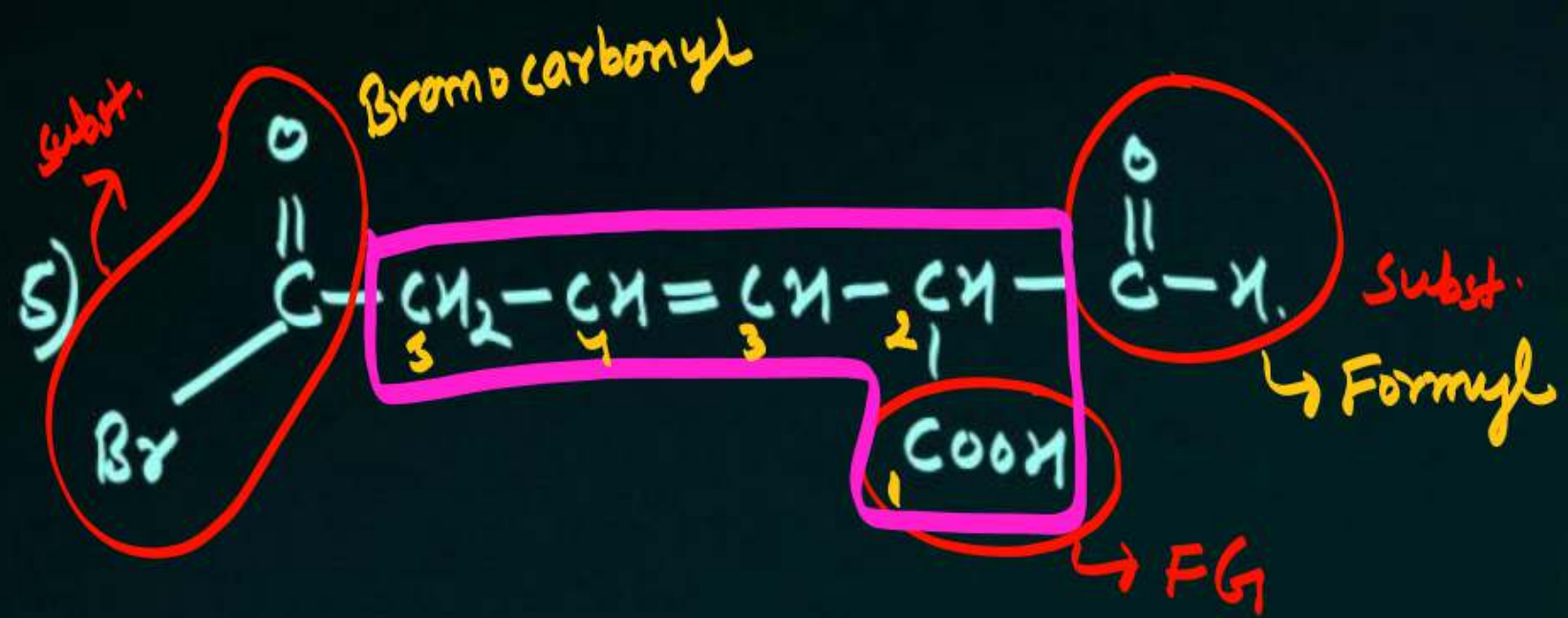
— OH ✓

— NH₂ ✓

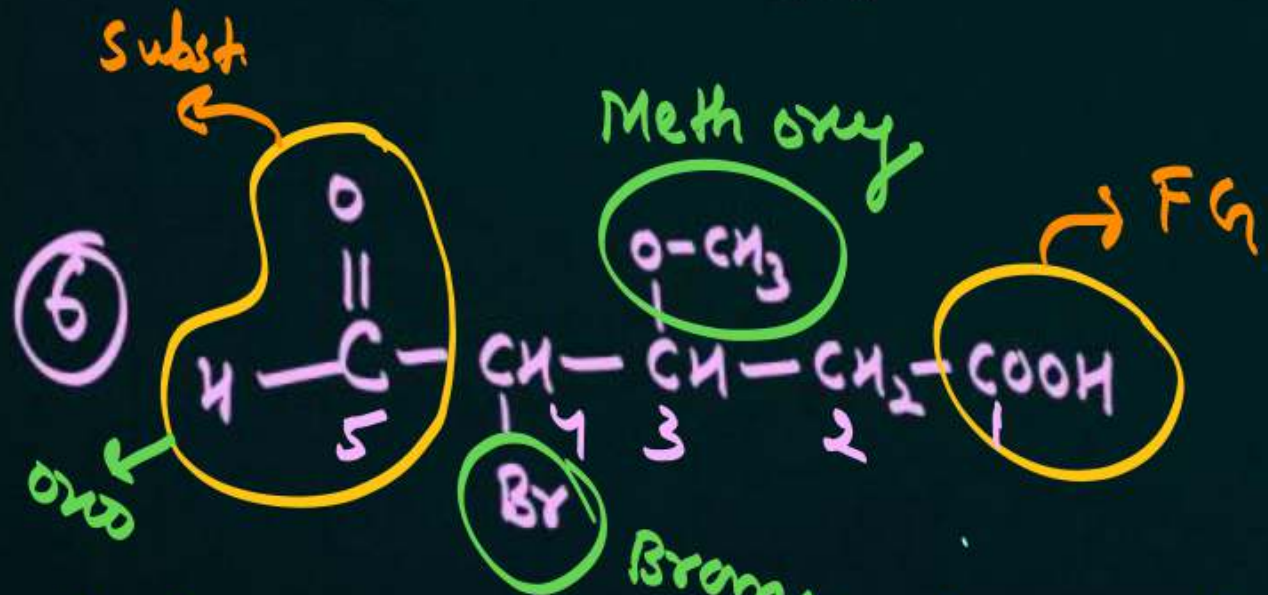


$\text{PCC} - \text{CN}$ (cyano).

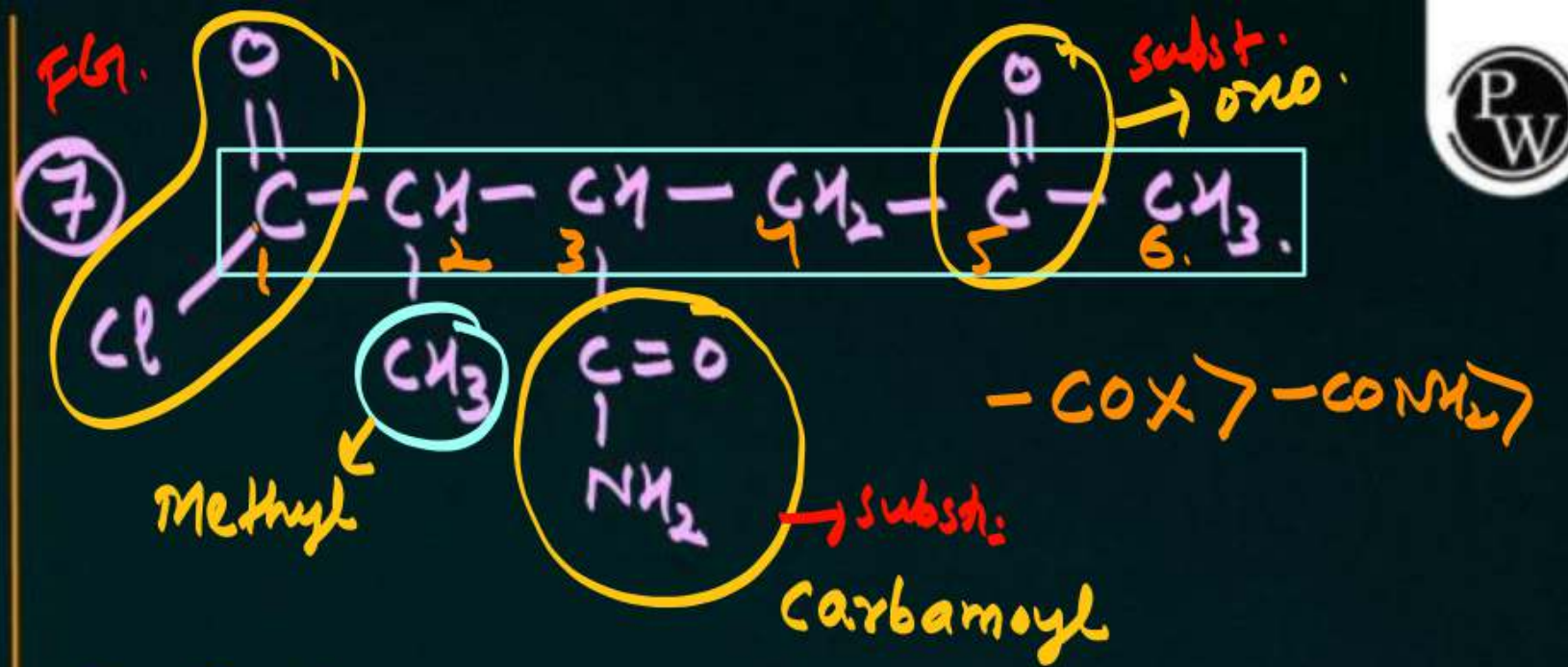
$\text{PCC} \rightarrow \text{NC}$ (Isocyano).



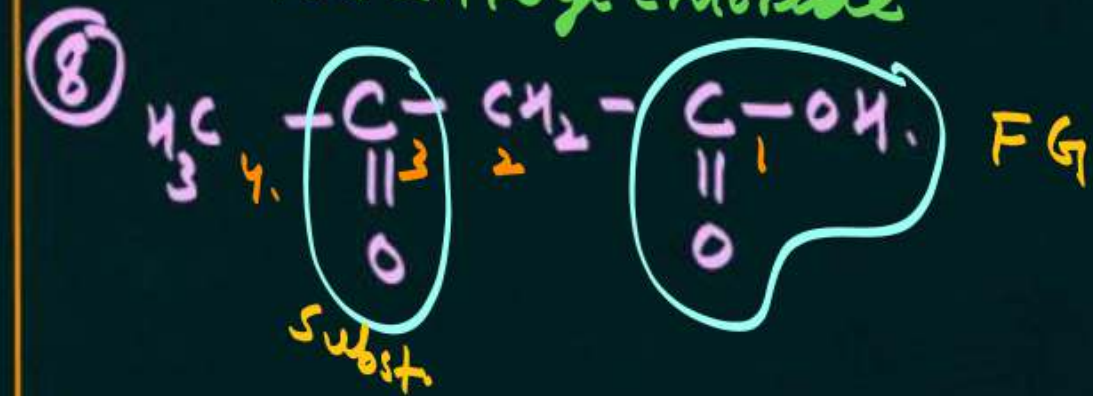
5-Bromocarbonyl-2-formyl pent-3-en-1-oic acid.



4-Bromo-3-methoxy-5-oxopentan-1-oic acid.



3-Carbamoyl-2-methyl-5-oxohexan-1-oyl chloride



3-oxobutan-1-oic acid.

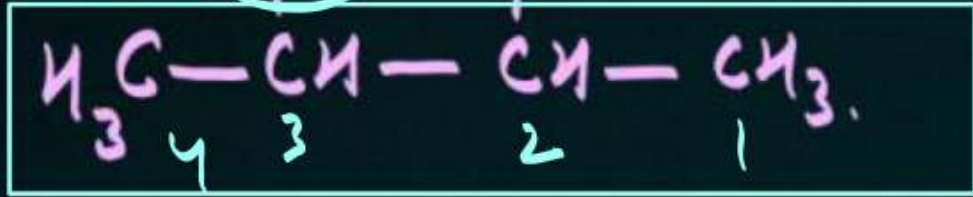
Subst (Amino)

FG

NH₂

OH

⑨



Thank
You